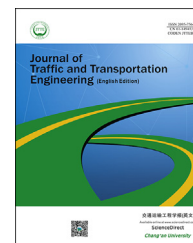


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Original Research Paper

Parking detection method using quadtree decomposition analysis

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HIGHLIGHTS

- A new method to detect available parking spaces in outdoor parking was proposed.
- The proposed method uses grayscale images obtained from any type of video camera.
- Using real-life data, the proposed method achieved a detection rate of 99.7%.
- The achieved detection rate is higher than most available methods in the literature.

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ABSTRACT

Searching for available parking spaces can be a painful experience for drivers due to driving around until finding a vacant space. This study proposes a new method to automatically detect available parking spaces. The proposed system identifies empty parking spaces using grayscale images obtained from any type of video camera. The method was found to successfully identify parking availability under different conditions and scenarios. The method was tested using real-life data and achieved a detection rate of 99.7%. This method can be applied in real-time to monitor parking availability and guide drivers to empty spaces. The method has several advantages, including simple algorithms, the use of low-quality black and white images, and simple configuration. Therefore, the system can provide enormous cost savings for locations with existing black and white surveillance cameras instead of replacing existing cameras with new high-quality cameras.

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1. Introduction

Parking problems play a major role in any transport system and have been one of the main concerns in most cities

(Ottomanelli et al., 2011; Shaaban et al., 2021). Searching and finding parking spaces is a problem for many drivers and can be a painful experience due to driving around to empty spaces (Bai et al., 2004; Golias et al., 2002). This search leads to an increase in fuel use, cost, and air pollution (Künzli

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et al., 2000; Shaaban and Pande, 2016). It also can be dangerous due to fast and sometimes risk maneuvers as soon as they find a parking space. The problems related to parking accessibility and the satisfaction of the drivers are major problems when it comes to the development of smart cities (Arellano-Verdejo et al., 2019). For smart cities, parking is considered one of the essential infrastructure elements (Atta et al., 2020; Perković et al., 2020). To solve this problem, several studies have focused on the allocation, availability, and layout of parking within the city (Parmar et al., 2020; Shen et al., 2019; Tu et al., 2020; Wang et al., 2007; Zhang et al., 2019). Other studies focused on parking policies and regulations (Gabbe et al., 2020; Ostermeijer et al., 2019; Taylor, 2020). Another line of research focused on identifying methods to determine the availability of vacant spaces then direct users to these spaces can save time, money, and effort (Siddiqui et al., 2020).

Automated parking identification methods can play a major in achieving this goal, which can be useful for drivers, parking lot owners, and the environment (Chinrungrueng et al., 2007; Sarangi et al., 2019; Shaheen et al., 2005). Parking detection methods can either determine the occupancy of an entire parking lot or identify the occupancy of every parking space. The objective of this research is to develop a new parking space detection method that can detect the occupancy of parking spaces. This paper presents an extension of our previous work which proposed two other parking detection methods as discussed in the literature review (Shaaban and Tounsi, 2015). There are currently several detection methods available in the literature. However, there is always a need for less expensive, effective, accurate, and easy to implement detection methods. The method proposed in this study needs a limited number of video cameras and in some cases only one video camera depending on the parking lot area, configuration, and size.

In this case, the total cost, including initial installation and regular maintenance is much less than other methods. Moreover, such a method will require a lower number of processing units to collect, analyze, and transfer the parking occupancy data. Finally, a detection method should provide accurate information to provide time and cost savings to the parking lot owners and drivers in addition to benefiting the environment by reducing the travel time, emissions, and fuel consumption for vehicles searching for parking spaces.

2. Literature review

Different studies identified methods that use image processing for the detection of vehicles. Some methods depended on fixed video cameras. Yamada and Mizuno developed a method to detect vehicles in parking lots using videos collected from high locations in the parking. The pictures were transformed to grayscale then used to recognize the vehicles' presence. A parking space was identified as occupied if a large number of small segments was identified and vice versa. The method was verified using actual outdoor parking data. The detection rate was found to be 98.7%. The mis-detections were due to camera dip, the insufficient difference in density with the surface of the parking, or flare reflections

from the ground surface during sunny times (Yamada and Mizuno, 2001).

True proposed a method that depends on computer vision algorithms for the detection of available parking spaces. A combination of colored histograms and vehicle feature detection was used. The system involved four stages including extraction of images, creation of colored histograms, feature detection for vehicles, and final classification. The first algorithm, support vector machine algorithm, indicated a detection rate of 94%. The second algorithm, *k*-nearest neighbor algorithm, showed a detection rate of 89% (True, 2007). Ichihashi et al. introduced an image-based method to detect parking occupancy using an algorithm based on fuzzy c-means clustering. The proposed method involved different stages including extraction of digital images from the video clips collected, using a sample for training, then testing new images using the developed algorithm. The proposed method produced a detection rate of 99.4% (Ichihashi et al., 2009).

Other methods utilized moving vehicles in the detection process. Hirahara and Ikeuchi (2003) developed two detection methods that used images collected from a laser range finder and a line scan camera installed in a moving vehicle. The detection rate for the laser range finder was 76.9%. For the camera, two methods of cluster analysis were used. One method was based on clustering points obtained from each scan. The second method was based on clustering points from several scans. A cluster of range points indicated the presence of a vehicle. The second method performed better than the first point with a detection rate that exceeded 90%.

Yu and Chen (2009) developed an algorithm that is based on videos collected while driving. A video camera is installed in a vehicle in the direction of the sidewalk. The vehicle moves inside the parking lot with an almost fixed speed of 15 mph or less. The process included four major stages. In the pre-processing stage, a matching method is applied to pairs of consecutive frames, to generate a disparity estimation for a sample of data. This sample is then added to the training library to be used in the classification stage. Two classification methods were used; the nearest neighbor and the Gaussian model were applied. In the last stage, a score was generated for each frame to indicate the parking space condition. The detection rate for both methods was above 99%.

Suhr et al. (2010) developed a method for parking space detection using motion stereo-based 3D reconstruction. Images were collected using one rearview fisheye camera. A three-dimensional model was reconstructed to determine the position of the vehicles in the images then locate the parking spaces available. The proposed method was compared with data acquired using a laser scanner and resulted in a detection rate of 90.3%.

In our previous work, two algorithms were assessed. The first algorithm used the maximum value of the image histogram, and the second algorithm used the bandwidth of the image histogram. Both algorithms were able to effectively recognize vacant and occupied parking spaces for different cases. The detection rate for both algorithms exceeded 98% for both algorithms (Shaaban and Tounsi, 2015).

In summary, some methods used a moving vehicle, which is time and effort consuming. It also needs a designated driver

to capture the data regularly (Hirahara and Ikeuchi, 2003; Suhr et al., 2010; Yu and Chen, 2009). Other methods depend on image processing. However, they require a long preparation process in the form of pre-processing, training, and building a library of references for comparison purposes with new images. This long process results in a long processing time for classification and comparison purposes. It also needs to keep a database in the form of the library for future comparison purposes (Ichihashi et al., 2009; True, 2007). One

study tested the quadtree technique. However, the study indicated that this method requires high-quality images and is not effective in the case of high noise levels in the image (Liu, 2005).

In this study, a new method using a quadtree technique is developed. The proposed method can be used in outdoor parking lots and does not require high-quality images. In this case, less expensive video cameras or existing surveillance cameras can be used proving tremendous cost savings. The

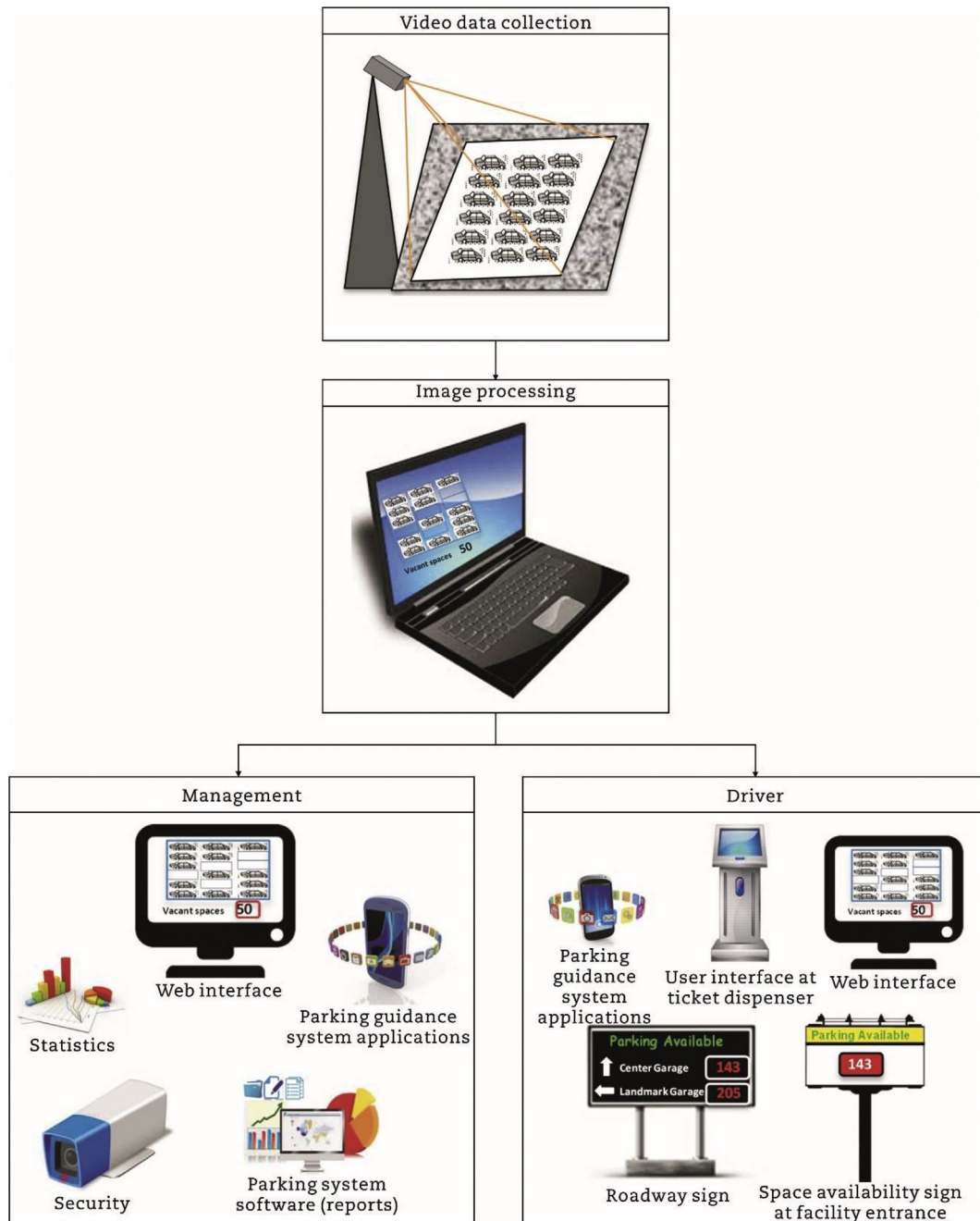


Fig. 1 – Architecture of the proposed system.

proposed method does not require path tracking or reference images. Therefore, there is no necessity for a long pre-processing preparation, training phase, or a library of references.

3. Method

3.1. System architecture

The proposed method depends on images obtained from video cameras fixed in a high position such as a high pole or wall to have coverage for the area needed. To start the processing, videos obtained are sent to a central computer. The proposed method is then applied to provide the number and location of available parking spaces in real-time. This information is provided to the drivers using different methods such as nearby dynamic message signs or at the entrance, a mobile application, a user interface at the ticket dispenser, a web interface online, etc. Then, drivers can obtain real-time parking lot information and make reservations if needed. On the other hand, garage operators can use the data for statistics regarding the use of the garage. The architecture of the proposed system is presented in Fig. 1.

3.2. Principle of detection

The vehicles are in the form of different shapes, sizes, and colors compared to white mark-off lines, casual shadows, or puddles for the parking spaces. Therefore, the detection depends on the common features of vehicles to detect their presence. The different elements of a vehicle include the hood, windows, headlights, bumpers, front grille, and license plate. The image taken for a vehicle will show the properties of the vehicles such as color, density, and texture. In concept, a parking space is considered occupied if it has a high number of segments. If not, the parking space is considered vacant.

In this study, a quadtree segmentation (grayscale level) method is used to analyze each space. Assuming that every car has various components, which will play a role in the subdivision. On the other hand, vacant spaces would not have these characteristics, even in the case of having white lines, building shadows, or puddles. Based on the above reasoning, the following solution is proposed. The obtained parking spaces' images are converted to grayscale level, and space is recognized as occupied or vacant based on the amount of division that has been executed.

3.3. System modules

Frames are obtained from the videos and converted to a grayscale level. Each frame is then compared to the previous frame, to identify if there are any noticeable changes. If no noticeable change appeared, the first frame is discarded and the count of the vacant parking spaces remains the same. Otherwise, if a noticeable change is shown, the frame is resized to a square image (512 pixel $\times$ 512 pixel) then the quadtree algorithm is applied. Based on the algorithm, images of parking cells are segmented by gray level as explained in the flowchart shown in Fig. 2. The same process is repeated iteratively until the criterion is met for each block meets. As

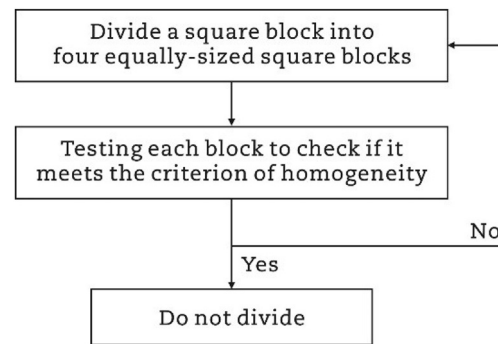


Fig. 2 – Quadtree algorithm flowchart.

a result, an image subdivided into several blocks can be obtained. A threshold on the number of blocks is established experimentally.

3.4. Image preparation and grayscale conversion

Images are converted from a true-color RGB image to grayscale. MATLAB functions are used for the conversion process by creating a weighted sum of the R, G, and B components: $0.2989R + 0.5870G + 0.1140B$. First, the RGB values are converted to NTSC coordinates. Second, the hue and saturation components are set to zero then converted back to RGB color space. Fig. 3 denotes the RGB to grayscale intensities conversion. For an 8-bit grayscale image, there are 256 different possible intensities, and so the histogram of the

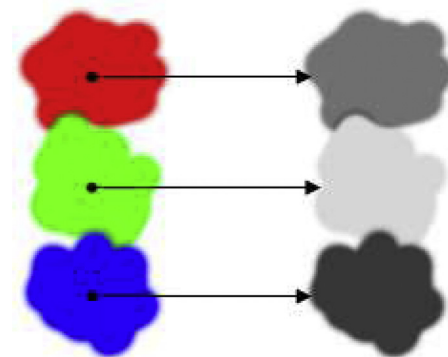


Fig. 3 – RGB conversion to grayscale.

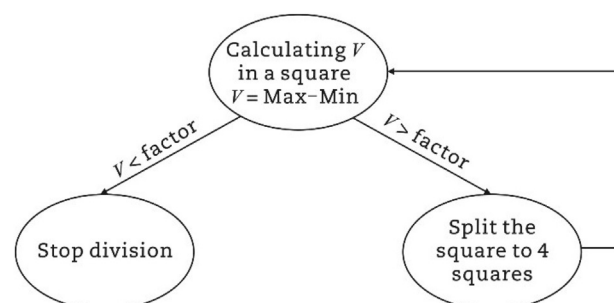


Fig. 4 – Criterion of homogeneity calculations.

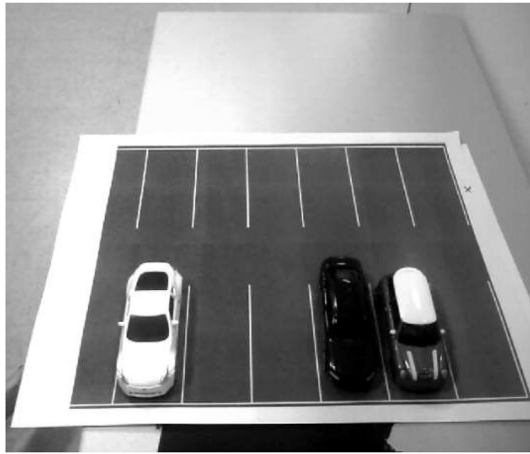


Fig. 5 – Mock-up experiment.

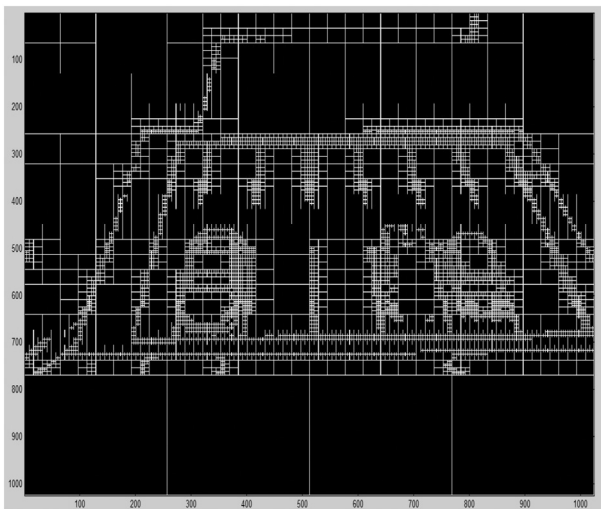


Fig. 6 – Quadtree results for a sample of occupied cells.

grayscale will graphically display 256 bins showing the distribution of pixels amongst those grayscale values. A true-color image can be converted to a grayscale image by preserving the luminance (brightness) of the image. The resulting image will be two-dimensional. The value 0 represents black and the value 255 represents white. The range will be between black and white values. The MATLAB function used is `rgb2gray`. Once applied $I = \text{rgb2gray}(RGB)$, it converts the true color image RGB to the grayscale intensity image I as shown in Fig. 3.

3.5. Quadtree concept

The main goal of image segmentation is partitioning the domain of an image into a set of disjoint regions. These regions are visually different and homogeneous with respect to some characteristics or computed properties. Those properties include grey level, texture, or color to enable easy image analysis. As a result, an image will be more meaningful and easier to analyze. Image segmentation is typically used to locate objects and boundaries (lines, curves, etc.) in images. In this method, this technique is used to classify the parking space for the presence of vehicles.

This decomposition splits an image into four equal square blocks. Each block is then tested to identify if it meets some criterion of homogeneity. If the criterion is met for a block, this block will not be divided anymore. If the criterion is not met, the block is split again into four blocks, and the test criterion is repeated for the new blocks. The same process is applied until each block reaches the criterion. The result can have blocks of several different sizes. As shown in Fig. 4, the criterion of homogeneity is calculated as the difference between the maximum and minimum pixel value in a square.

In Fig. 4, Max is the maximum value in the square block, Min is the minimum value in the square block, factor is the threshold defined for the division. The less the value is the more blocks we got.

The function used in the study is “qtdecomp” (Eq. (1)).



Fig. 7 – A screenshot of the study area.

$$S = \text{qtdecomp}(\text{Img}, \text{threshold}) \quad (1)$$

where *Img* is a grayscale square image, *threshold* is the factor of division. It is specified as a value that varies from 0 to 1.

The $S = \text{qtdecomp}(\text{Img}, \text{threshold})$ function splits a block if the maximum value of the block elements minus the minimum value of the block elements is greater than the threshold. The threshold is specified as a value between 0 and 1, even if *Img* is of class *uint8* or *uint16*. If *Img* is *uint8*, 255 to determine the actual threshold to use multiply the threshold value you supply; if it is *uint16*, the threshold value supplied is multiplied by 65,535.

The threshold on the number of blocks is established experimentally. Identifying the threshold is conducted by applying the algorithm for 100 slots then calculating the percentage of segments compared to the number of the whole pixels in the slot. This method has several advantages including quick processing, easily finding areas of interest, simply reducing or increasing resolution by dealing with nodes, and not needing a database. The main disadvantage of this method is that it could suffer if the image has a high noise level.

3.6. Mock-up experiment

A mock-up experiment was conducted in the lab using a drafting board and toy cars as shown in Fig. 5. A mock-up is a prototype if it provides at least part of the functionality of a system and enables testing of a design. The physical mock-ups are often a physical prototype of an environment, built to test design decisions. These decisions can be based on input from highly integrated teams of end-users, designers, managers, and researchers. Physical mock-ups can have a high level of fidelity, interaction, and user immersion. The level of fidelity in a physical mock-up can vary among simple, detailed, or live (i.e., functional). Simple mock-ups can consist of tape on the floor, cardboard boxes, dry erase boards, and foam-core walls.

Before applying the proposed approach in real data, it was necessary to test the algorithm. The purpose of the experiment is to simulate the parking lot with vacant and occupied spaces using toy cars in controlled conditions. In this experiment, toy cars were joined with a background paddle painted as parking ground. The combination of 12 parking spots and six different cars was sufficient to simulate a similar concept of real parking. A video camera was mounted on the top of the screen and oriented to a group of toys with printed parking. These miniatures were used to enable an in-lab parking mock-up which was used to test and validate different parking configurations to confirm the algorithm. MATLAB was used for image conversion and image detection using the quadtree algorithm.

Images of parking cells taken from the mock-up were segmented by gray level using quadtree decomposition to divide parking spaces into small different-sized squares. To check if the selected place in the step above is empty or not, the number of segments in each cell was counted as presented by the white pixels in Fig. 6. After deciding a threshold, it can be decided whether a cell is occupied or vacant. In this step, the threshold value is decided. One threshold value is for the detection of the spaces because it is in the form of percentage

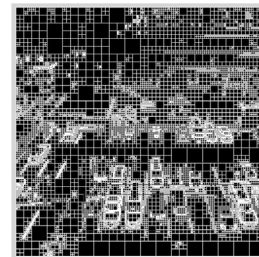
thus, although slots' sizes are not equal, detection can be done by only one threshold.

As part of the experiment, the positioning of cars varied. Also, a uniform dark-colored car was used to test if it will cause problems in segmentation. However, in all cases, the car presence resulted in white segments, which represent the shapes of those cars. The number of white pixels was radically different between an occupied place and an empty one. This confirms that quadtree is a suitable method to define the presence of an object.

3.7. Data collection

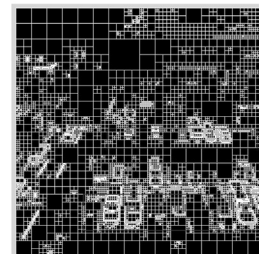
To test the proposed method using real-life data, data were collected from an outdoor parking lot in the City of Vancouver, Canada as shown in Fig. 7. A VGA web camera was mounted in a high position. The web camera ought to have almost the same

(a)



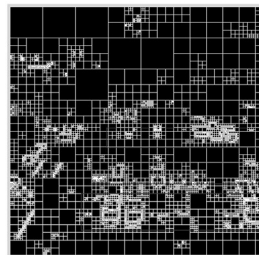
Row	White pixel										
1	484	542	314	552	301	567	580	699	767	433	327
2	874	1059	653	989	829	705	546	662	992		
3	1590	2169	2004	1406	1461	1246	1339	1913			

(b)



Row	White pixel										
1	314	398	232	461	242	420	550	649	718	342	288
2	692	883	474	861	609	560	381	482	853		
3	1264	1913	1790	1099	1083	846	1012	1628			

(c)



Row	White pixel										
1	248	322	199	346	173	258	507	590	666	206	205
2	598	762	365	703	522	510	196	416	756		
3	969	1693	1567	832	861	669	773	1402			

Fig. 8 – A sample of quadtree segmentation results. (a) Factor = 0.10. (b) Factor = 0.15. (c) Factor = 0.20.

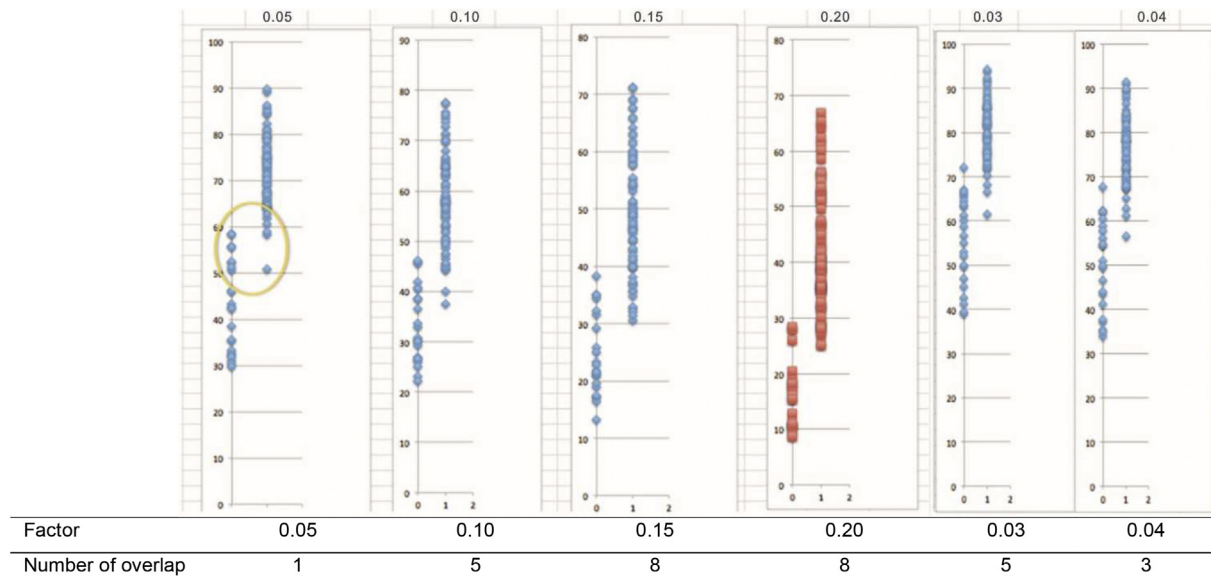


Fig. 9 – Scatter diagrams for different factors.

features as the surveillance cameras—perhaps worse—in terms of resolution and image types and this was challenging but ensuring. The camera was mounted in the room which was about 30 m above the ground with a dip of 30°. The shooting range was about 25 m wide by 40 m deep, with 28 parking spaces involved. The parking of the hotel contained three rows. The first row contained 11 spaces, the second row had 9 spaces and 8 spaces composed the third row. The number of pixels for each space in images was different based on each row of parking as each row contained similar size spaces. Based on the recording time of 8 h, there were 32 frames (one frame every 15 min) and 28 spaces per frame, which results in 896 frames. Rainy and cloudy weather was selected as a preferred situation where the brightness and clearness of the images would suffer. It was chosen intentionally to guarantee a good performance of the algorithm with bad conditions assuring that it would definitively work for normal weather.

4. Analysis

Computations were performed with a program written in MATLAB. Image Processing Toolbox, which provides a comprehensive set of reference-standard algorithms, functions, and applications for image processing, analysis, visualization, and algorithm development. It helped to perform image analysis, image segmentation, image enhancement noise reduction, geometric transformations, and image registration. Similarly, the visualization functions and applications explored images and videos and examined a region of pixels, adjust color and contrast, create contours or histograms, and manipulated regions of interest.

The first 10 frames (280 individual frames) were used to determine the optimal threshold. The quadtree method was applied to the 280 spaces using four different factors (0.05,

0.10, 0.15, and 0.20). The predefined MATLAB function was used to apply the quadtree segmentation. The function splits a block if the maximum value of the block elements minus the minimum value of the block elements is greater than the threshold. A sample of the results is shown in Fig. 8. The numbers represent the white pixels inside a spot. More white pixels indicate more details in the parking space. The bold numbers mean that they are below the threshold, which means they are detected as empty places.

For each parking space, the percentage of white pixels was extracted and compared to the whole size of the slot. Scatter diagrams were drawn of the percentages to conclude the best factor to use as shown in Fig. 9. The best factor should provide almost two separate groups presenting vacant and non-vacant places. When comparing the initial four studied factors (0.05, 0.10, 0.15, and 0.20), it was found that the 0.05 factor provided the lowest number of overlaps (only 1 overlap) compared to 5 for 0.10, 8 for 0.15, and 8 for 0.20. To identify if there is a better factor than 0.05, two additional factors (0.03 and 0.04) were tested. Both provided a higher overlap (5 and 3) than the 0.05 factor. Therefore, a factor of 0.05 was used for the study case.

For the case study, the results indicated that 0.05 is the best division factor value, which indicated the best detection. Applying this factor to all data (896 frames), an error of only 0.3% was obtained with a detection rate of 99.7%. The error was decided based on performing a manual check to determine whether the detected parking space was vacant or occupied. The highest number of missed vehicles in any frame was one vehicle. In addition, the highest percentage of error in one frame was 4.8% (one out of 21 occupied vehicles was not detected). Also, the total number of frames with an error in identifying occupied parking spaces was 55 frames (6.1%). Thus, the proposed methods of vehicle detection were found to be efficient. The quadtree process took 0.051627 s. The quadtree process plus the detection

Table 1 – Comparison with related parking detection studies.

Study	Data	Facility	Image	Detection rate (%)
Yamada and Mizuno (2001)	Grayscale image	Outdoor parking	Fixed video camera	98.7
True (2007)	Color image	Outdoor parking	Fixed video camera	91.0*
Ichihashi et al. (2009)	Color image	Outdoor parking	Fixed video camera	99.4
Shaaban and Tounsi (2015)	Grayscale image	Outdoor parking	Fixed video camera	99.2*
Koushika et al. (2021)	Grayscale image	Outdoor parking	Fixed video camera	94.0
Bibi et al. (2017)	Grayscale image	Outdoor parking	Fixed video camera	98.0*
Iqbal et al. (2021)	Color image	Outdoor parking	Fixed video camera	97.6
Yu and Chen (2009)	Grayscale image	Outdoor parking	Moving video camera	99.5*
Suhr et al. (2010)	Grayscale image	Outdoor parking	Moving video camera	90.3
This study	Grayscale Image	Outdoor parking	Fixed video camera	99.7

Note: * Multiple methods were used. Only the highest detection rate is shown.

took 0.092403 s. It should be noted that a MacBook Pro with Processor 2.8 GHz Intel Core i7, Memory 4 GB 1333 MHz DDR3 was used.

5. Conclusion

The population growth in most cities requires the design of efficient and sustainable traffic systems by taking full advantage of modern-day technology (Ata et al., 2019). Smart parking systems are one of the main components needed for the infrastructure of smart cities (Iqbal et al., 2018). The study proposed a new method for the detection of available parking spaces in outdoor parking lots using grayscale images acquired from video cameras. The proposed method is simple and accurately identified vacant parking spaces under different conditions. Using real-life data, the proposed method achieved a detection rate of 99.7%. The achieved detection rate is higher than most available methods in the literature as shown in Table 1.

These results can be enhanced if better quality cameras are used. Future studies could focus on testing the proposed method for other types of facilities to verify the detection rate such as large parking lots, indoor parking, parallel parking, and night conditions. Finally, developing a smart system that is based on video detection obtained from fixed surveillance video cameras in addition to other methods that depend on parking surveillance and enforcement technologies such as drones (Arun et al., 2021; D'Aloia et al., 2015; Maria et al., 2016; Sarkar et al., 2019) and dashboard cameras in parking enforcement vehicles (Kumar et al., 2021; Li et al., 2021; Matsuda et al., 2021a, b) should be investigated.

Conflict of interest

The authors do not have any conflict of interest with other entities or researchers.

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